Bokinala **Samyak Kumar**

Undergraduate Researcher · Marine Robotics

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PROFESSIONAL SUMMARY

Undergraduate researcher in Marine Robotics at the Indian Institute of Technology Madras, working on Autonomous Underwater Vehicles (AUVs), underwater perception, and autonomous navigation. Research interests span sonar-based perception, computer vision, sensor integrity monitoring, robot learning, and autonomous docking. Experienced in developing sensor-rich AUV platforms through end-to-end integration of perception, navigation, control, and embedded systems. Currently contributing to research in sonar-based perception and information-driven active search for autonomous docking of underwater vehicles.

EDUCATION

B.Tech in Naval Architecture and Ocean Engineering

Expected July 2027

Indian Institute of Technology Madras

- **NIRF Rank 1 Engineering Institute in India**
- **Coursework:** Marine Robotics, Marine Instrumentation, Maneuvering and Control of Marine Vehicles, Ship Hydrodynamics, Machine Learning for Ocean Engineers, Ship Design, Design of Ocean Structures, Marine Engineering, Guidance and Control of Marine Vehicles

PUBLICATIONS

Deep Learning Based Sensor Integrity Monitoring for Robust AUV Navigation

Bokinala Samyak Kumar, Vallabh Deogaonkar, Mohammed Ibrahim M., and Abhilash Somayajula.

*Presented at **OCEANS Sanya 2026**, Sanya, China. IEEE Xplore*

- Developed a physics-based simulation framework based on nonlinear 6-DOF AUV dynamics for generating labeled multi-sensor datasets with realistic underwater degradation and fault conditions.
- Modeled mission-relevant anomalies affecting IMU, DVL, and pressure sensors, including inertial drift, bottom-lock degradation, acoustic disturbances, frozen outputs, and missing-data failures.
- Designed a multi-head LSTM-based temporal learning framework capable of distinguishing nominal operation, short-term degradation, and persistent faults from sequential sensor measurements.
- Introduced a scenario-level evaluation methodology comprising 20 representative combinations of sensor anomalies to assess mission-level robustness under realistic operating conditions.
- Demonstrated reliable real-time sensor integrity monitoring across diverse disturbance patterns, enabling improved fault awareness and navigation reliability for autonomous underwater vehicles.

3D Reconstruction of Underwater Structures using Forward-Scan Sonar and Gaussian Splatting

Vallabh Deogaonkar, Mohammed Ibrahim M., **Bokinala Samyak Kumar**, Aatmaj Bhayani, and Abhilash Somayajula.

*Presented at **OCEANS Sanya 2026**, Sanya, China. IEEE Xplore*

- Developed a Gaussian Splatting-based reconstruction framework for recovering three-dimensional underwater structures from sparse forward-scan sonar measurements.
- Formulated the elevation ambiguity problem inherent to forward-scan sonar and represented individual sonar observations as elevation-ambiguous acoustic arcs in three-dimensional space.
- Proposed multi-view consistency filtering and elevation-aware Gaussian initialization techniques to generate physically plausible scene representations from sequential sonar observations.
- Introduced view-dependent optimization, elevation-ray regularization, opacity-guided refinement, and reconstruction-aware pruning strategies to improve geometric consistency and suppress spurious structures.
- Demonstrated accurate reconstruction of complex underwater structures using low-cost acoustic sensing, highlighting the potential of sonar-only perception for underwater inspection, mapping, and autonomous marine robotics.

EXPERIENCE

Undergraduate Researcher

2024 – Present

MAVLAB, IIT Madras

- Developed a compact research-grade AUV as a platform for underwater perception, navigation and mapping research.
- Integrated DVL, IMU, pressure sensor and imaging, forward-scan and side-scan sonars into a single sensing architecture.
- Designed ROS 2 software for synchronised multi-sensor data acquisition, real-time visualisation, logging and analysis.
- Developing sensor fusion, state estimation and sonar processing pipelines for robust underwater navigation and mapping.

Team Captain and Autonomous Systems Lead

2025 – 2026

Team Aritra, IIT Madras

- Led the development, and field validation of a fully autonomous AUV for the Singapore AUV Challenge (SAUVC).
- Developed perception, navigation, and control pipelines using acoustic and vision-based sensing systems for autonomous mission execution.
- Integrated multi-sensor architectures comprising DVL, IMU, imaging sonar, pressure sensors, acoustic communication, and localization systems.
- Designed autonomous mechanisms for target acquisition, reacquisition, and mission-specific underwater tasks.

Mechatronics Design Engineer *2024 – 2025*

Team Amogh, IIT Madras

- Contributed to the development of an Autonomous Underwater Vehicle (AUV) for RoboSub and SAUVC international competitions.
- Designed electrical architectures including power distribution, PCB systems, sensing interfaces, and safety-critical subsystems.
- Developed PCB schematics and layouts using KiCad for reliable underwater robotic operation.
- Designed mechanical subsystems and internal vehicle layouts using Fusion 360, focusing on compact integration and maintainability.
- Performed hardware integration, fabrication, waterproofing, and full-system assembly of underwater robotic platforms.

AUV PLATFORMS

MAKO – Torpedo AUV for Autonomous Docking and Bathymetry *MAVLAB, IIT Madras*

- Designed the CAD and PCB architecture of a 1.4 m torpedo AUV with X-configured fins, and integrated IMU, DVL, depth sensor, dual side-scan sonars and a docking camera.
- Developed survey missions (lawnmower, orbit, spiral and depth-varying) with a virtual geofence, plus a desktop 3D viewer and mission editor.
- Developed vision-based docking: detection of the dock's guide lights, an EKF to estimate the dock pose, and a guidance state machine to search, approach and enter the dock.
- Built an offline replica of the lab simulator's dynamics to find and fix errors in its thrust, fin and added-mass models.

TIMI – Competition AUV for SAUVC *Team Aritra, IIT Madras*

- Developed the 6-DOF control stack in ROS 2 for 8 thrusters: cascaded position–velocity PID, attitude PD and thrust allocation.
- Implemented station keeping, waypoint and spline path following, and the competition mission logic (gate, drum detection, ball drop).

DHWANI – Acoustics-Focused AUV *MAVLAB, IIT Madras*

- Designed the CAD of a compact AUV for bathymetry and SLAM research, with a slide-out electronics frame for servicing.
- Designed the power architecture around a Jetson Orin Nano and Pixhawk, with separately switched subsystem sections.
- Integrated echosounder, forward-scan, side-scan and imaging sonars, USBL, DVL, IMU and pressure sensor on one AUV.
- Implemented depth, roll and yaw controllers and ran pool tests logging vehicle motion with sonar data for mapping.

PROJECTS

Information-Driven Active Search and Autonomous Docking of an Underactuated AUV *Ongoing, MAVLAB*

- Building an autonomous search-and-docking pipeline for an underactuated torpedo AUV with no prior dock location.
- Updating a belief over dock locations from side-scan sonar and choosing viewpoints by information gain vs. travel cost.
- Identifying the dock from an asymmetric light pattern, then homing with optimal control and camera-based visual servoing.
- Benchmarking against coverage-based search in simulation, then trials on the real AUV in a $30 \times 30 \times 3$ m wave basin.

AUV Guidance, Navigation & Control Platform (REMUS-100) *Python, Dash*

- Modelled the REMUS-100 nonlinear dynamics, linearised them, and analysed controllability and observability.
- Designed PID autopilots by pole placement, and LOS/ILOS path-following and proportional-navigation interception guidance.
- Implemented and compared Luenberger, Kalman and extended Kalman filters, including estimation of ocean currents.
- Extended it to multi-AUV area coverage with task assignment and lane reallocation when a vehicle fails.

Physics-Based Side-Scan Sonar Simulator *Python, Numba, PyQt5*

- Developed a ray-casting simulator of a dual side-scan sonar, extending the acoustic model from our OCEANS 2026 paper.
- Modelled acoustic shadows, layover, seabed backscatter, absorption, speckle and gain, and generated sonar mosaics.
- Derived an analytic shadow model and used it to validate the simulator's geometry.

Unsupervised Segmentation of Underwater Images *PyTorch, DINOv2*

- Studied training-free foreground segmentation of underwater images using frozen DINOv2 (ViT-B/14) patch features.
- Clustered patch features with k-means, choosing the number of clusters per image automatically by silhouette score.
- Separated objects from water with a border rule that needs no labels, and benchmarked it against an oracle on USOD.
- Proposed its use on an AUV as a low-cost "object ahead" cue fused with sonar, with real-time onboard inference next.

AUV Fin Design & Optimization Platform *Python, XFOIL, NSGA-II*

- Developed a tool that derives fin size and shape from a turning requirement using vehicle dynamics, airfoil data and finite-wing theory.

- Added structural and servo checks, sensitivity analysis and drag–mass optimisation, with every equation documented and traceable.

AUV Gate Detection & Pose Estimation *OpenCV, ROS 2*

- Developed a classical OpenCV pipeline that boosts contrast in murky pool water and detects the gate's vertical poles.
- Classified each frame as no pole, one pole or full gate without inventing missing poles, and tracked the gate over time.
- Estimated the gate's position and heading relative to the camera using PnP, reporting only estimates that pass checks.
- Deployed as a ROS 2 node on the onboard Jetson, with tools to replay recorded pool videos and tune detection settings.

TECHNICAL SKILLS

Software & Tools: Python, ROS 2, Gazebo, Linux, Git, Docker
Estimation & Control: Kalman Filter, Extended Kalman Filter, Sensor Fusion, PID Control, Path-Following Guidance, Thrust Allocation
Perception & Machine Learning: Computer Vision (OpenCV), Deep Learning (PyTorch), Time-Series Models (LSTM), Image Segmentation
Sonar Processing: Forward-Scan Sonar, Side-Scan Sonar, Imaging Sonar, Sonar Simulation, Sonar-Based 3D Reconstruction
Hardware Design: Fusion 360, KiCad, PCB Design, Power Distribution, Embedded Systems (Jetson, Pixhawk, Arduino)
Marine Sensors: DVL, IMU, Pressure Sensor, Echosounder, USBL, Stereo Camera, GNSS

AWARDS & ACHIEVEMENTS

- Recipient of the **MTS Dieter Family Scholarship (2025), US$1,500**, awarded by the Marine Technology Society for academic excellence and contributions to marine technology initiatives.

PROFESSIONAL DEVELOPMENT

- Presented a research paper at the **OCEANS 2026 Sanya Conference**, Sanya, China, to an international research audience.
- Selected participant of the **IEEE OES Summer School 2026**, Sanya, China, on marine robotics, autonomy and ocean technology.
- Attended the **OCEANS 2025 Great Lakes Conference**, Chicago, USA, engaging with global researchers and industry leaders.
- Attended the **International Conference in Ocean Engineering (ICOE) 2025**, Chennai, India, on marine robotics and offshore systems.

LEADERSHIP & PROFESSIONAL SERVICE

Game Master, Aquavision AUV Competition *2025 – Present*

IEEE OES IIT Madras Student Chapter

- Led the design and execution of two editions of Aquavision, an Autonomous Underwater Vehicle (AUV) competition focused on marine robotics and underwater autonomy.
- Developed competition tasks, technical rules, and evaluation criteria to simulate real-world underwater robotic challenges.
- Coordinated event operations and participant engagement, fostering hands-on learning in autonomous marine systems.

Joint Secretary *2024 – 2025*

Marine Technology Society (MTS) IIT Madras Student Section

- Directed the planning and execution of technical events, workshops, and outreach initiatives promoting marine technology and ocean engineering.
- Organized Techsym 2025, facilitating collaboration between students, researchers, industry professionals, and government organizations.
- Contributed to strengthening student engagement in marine robotics, autonomous systems, and emerging ocean technologies.

Student Ambassador *2024 – 2025*

Royal Institution of Naval Architects (RINA)

- Represent IIT Madras within the international RINA student ambassador network and support initiatives promoting naval architecture and marine engineering.
- Facilitate student participation in professional, technical, and networking opportunities within the global maritime community.

RESEARCH INTERESTS

- Polar Robotics
- Under-Ice Exploration
- Field Robotics
- Long-Endurance Missions
- Marine Bioacoustics
- Sensor Fusion
- Sonar Mapping
- Underwater Perception
- Active Search
- Autonomous Docking
- Deep-Sea Exploration
- Embodied AI
- Fault Detection
- Resilient Navigation
- AUV Design